Name: ______________________ **Class:** ________ **Reg Number:** _____

MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 1

Chemistry **8872/02**

Paper 2 **10 September 2012**

2 hours

Additional Materials: *Data Booklet*
Writing Papers

READ THESE INSTRUCTIONS FIRST

Write your name, class and register number in the spaces at the top of this page.

This booklet contains Section **A** and Section **B**.

<u>Section A : Pg 2 to 12</u>

Answer **all** questions in Section A in the spaces provided on the question paper. You are advised to spend about **1 h** on Section **A**.

<u>Section B : Pg 13 to 18</u>

Answer **2 out of 3** questions in Section B. You are advised to spend about **1 h** on Section **B**.

Hand in Section **B** ***separately*** from Section **A**.

Fasten your answers for Section B behind the given **Cover Page**. Detach the **Cover Page** from the last page behind this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

Examiner's Use		
Paper 1	**MCQ**	**/ 30 m**
		/ 33 %
Paper 2 Section A	**Q1**	**/ 14 m**
	Q2	**/ 13 m**
	Q3	**/ 13 m**
Paper 2 Section B		**/ 40 m**
Paper 2 Total		**/ 80 m**
		/ 67 %
Grand Total		**/ 100 %**
Grade		

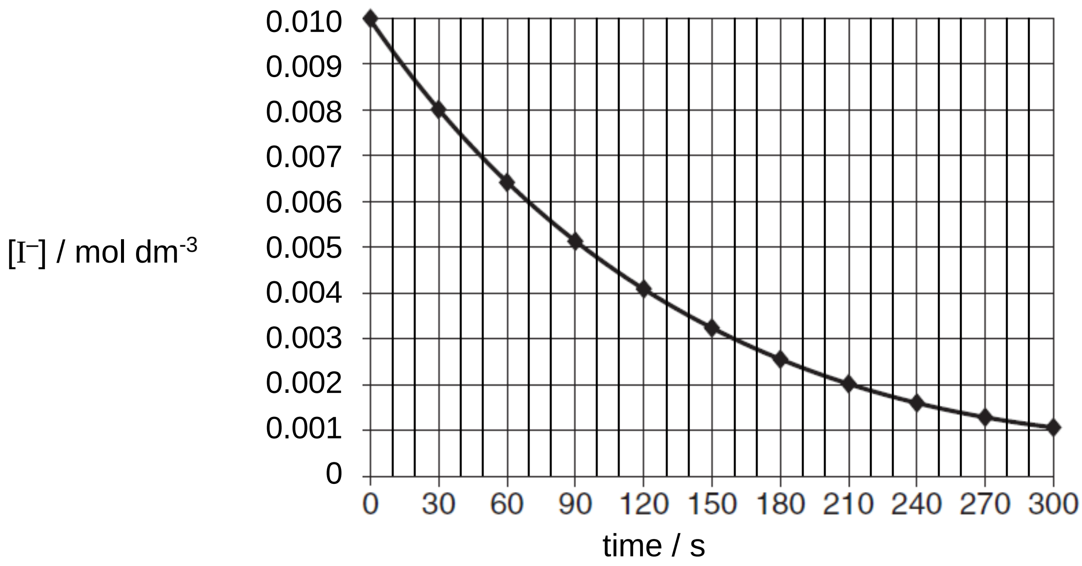
Section A : Structured Questions

Answer all the questions in this section in the spaces provided.

1 In the late 19th century, the two pioneers of the study of reaction kinetics, Vernon Harcourt and William Esson, studied the rate of the reaction between hydrogen peroxide and iodide ions in acidic solution.

$$H_2O_2 + 2I^- + 2H^+ \rightarrow 2H_2O + I_2$$

A study was conducted in which both $[H_2O_2]$ and $[H^+]$ were kept constant at 0.050 mol dm^{-3}, and $[I^-]$ was plotted against time. The following curve was obtained.



(a) Calculate the initial rate of this reaction, including its units.

[1]

The order of reaction with respect to I^- is one.

(b) Using the graph above, verify that the reaction is first order with respect to I^-.

……………………………………………………………………………………………………..

…………………………………………………………………………………………………….[1]

(c) To find out the orders of reaction with respect to H_2O_2 and H^+, various volumes of these solutions were mixed and the time taken for the colour of iodine to first appear, ***t***, was recorded. The results are shown in the table below.

Use the following data to deduce the orders with respect to $[H_2O_2]$ and $[H^+]$, explaining your reasoning.

Experiment	**$[H_2O_2]$ / mol dm^{-3}**	**$[I^-]$ / mol dm^{-3}**	**$[H^+]$ / mol dm^{-3}**	**Time taken for colour of iodine to first appear, *t* / s**
1	0.050	0.010	0.050	16
2	0.010	0.020	0.050	40
3	0.010	0.020	0.010	40

[3]

(d) Using your answers in **(a)** to **(c)**,

(i) write the rate equation for the reaction.

…………………………………………………………………………………………...

(ii) calculate the rate constant, *k*, including its units.

[2]

(e) Hydrogen peroxide can undergo decomposition to form oxygen and water. The rate of this reaction can be increased by the addition of a small amount of solid manganese(IV) oxide.

Explain, including a sketch of the Maxwell-Boltzmann curve, how the addition of solid manganese(IV) oxide can increase the rate of the decomposition of hydrogen peroxide.

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(f) **Reaction I**

50.0 cm^3 of an aqueous solution containing 0.10 mol dm^{-3} of hydrogen peroxide was added to 40.00 cm^3 of 0.10 mol dm^{-3} acidified sodium bromide.

Reaction II

Excess potassium iodide was then added to the solution from **Reaction I**.

The hydrogen peroxide which was left over from **Reaction I** liberated enough iodine to react with exactly 20.00 cm^3 of a 0.10 mol dm^{-3} of the sodium thiosulfate solution forming sodium tetrathionate, $Na_2S_4O_6$, and iodide ions.

$$H_2O_2 + 2I^- + 2H^+ \rightarrow 2H_2O + I_2$$

$$2NaS_2O_3 + I_2 \rightarrow NaS_4O_6 + 2I^-$$

(i) Calculate the number of moles of iodine produced in **Reaction II**.

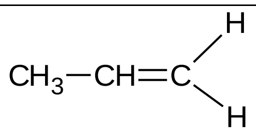
(ii) Hence, using your answers in **(f)(i)**, calculate the number of moles of bromide ions which reacted with one mole of hydrogen peroxide in **Reaction I**.

[4]

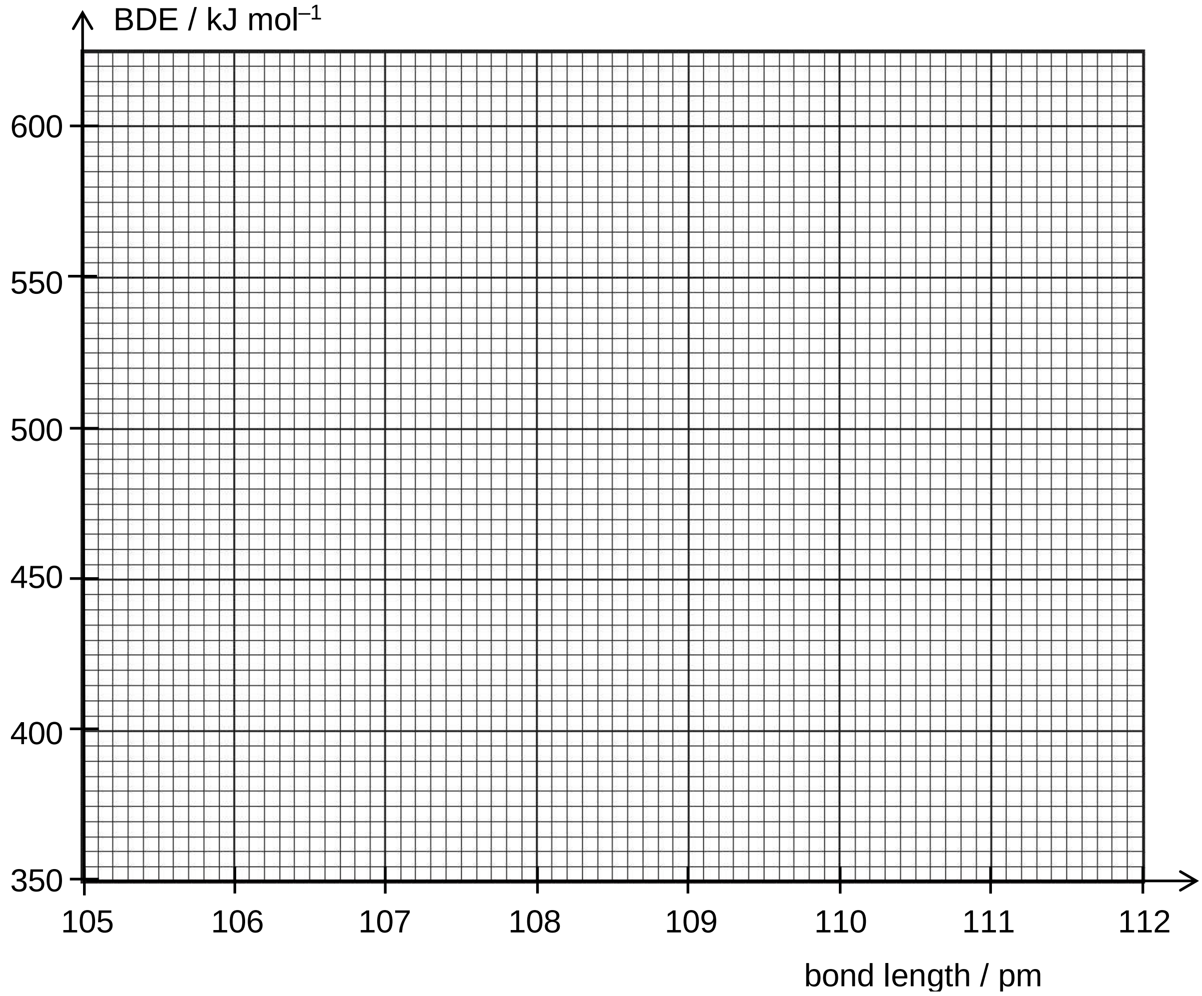
[Total: 14]

2 Bond energy values in the *Data Booklet* represent the **average** energy required to break one mole of a bond type between two atoms in its gaseous molecules. For example, the C–H bond energy value in the *Data Booklet* is 410 kJ mol^{-1}. This is an average of many different C–H bonds in various compounds. However, the **exact** value required to break one mole of a **specific bond** between two atoms in its gaseous molecules is termed bond dissociation energy, BDE.

The BDE values of some C–H bonds are presented in the table below.

Bond	**Bond dissociation energy, BDE / kJ mol^{-1}**	**Bond length / pm (1 pm = 10^{-12} m)**
ethane, CH_3CH_3	425	109.3
ethene, $CH_2{=}CH_2$	460	108.5
ethyne, $CH{\equiv}CH$	555	105.9
propene, 	375	110.6

(a) (i) On the axes below, plot the graph of BDE against bond length for the various C–H bonds given in the table above. Extrapolate your graph to intersect both axes.



(ii) In terms of orbitals overlap, briefly explain the relationship between BDE and bond length of C–H bond.

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…………………………………………………………………………..………...[2]

(b) Methylbenzene and bromomethylbenzene are liquids with boiling points of 111°C and 198°C respectively. Methylbenzene can react with bromine to form bromomethylbenzene.

$$C_6H_5CH_3(l) + Br_2(l) \rightarrow C_6H_5CH_2Br(l) + HBr(g)$$

(i) Calculate the standard enthalpy change of the reaction shown above using the following data.

Standard enthalpy change of formation of $C_6H_5CH_3(l)$	= +12 kJ mol^{-1}
Standard enthalpy change of formation of $C_6H_5CH_2Br(l)$	= –14 kJ mol^{-1}
Standard enthalpy change of formation of HBr(g)	= –36 kJ mol^{-1}

(ii) The BDE of C–Br bond in bromomethylbenzene is 244 kJ mol^{-1}.

Use this given data, your answer from **(b)(i)** and additional data from the *Data Booklet*, calculate the BDE of the C–H bond of the methyl group in methylbenzene. (You should **only** break and form the bonds which take part in this reaction.)

(iii) Use your graph in **(a)(i)** to predict the C–H bond length of the methyl group in methylbenzene. State your answer to **one** decimal place.

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(iv) Suggest why the BDE you have calculated in **(b)(iii)** differs from the value in the *Data Booklet*.

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…...…………………………………………………………………………………...[5]

(c) **(i)** Some bond energy values from *the Data Booklet* are given in the table below.

bond	**bond length / pm**	**bond energy / kJ mol^{-1}**
C–O	143	350
C–N	147	305

The C–O and C–N bond length are similar. However, the bond energies differ significantly. Suggest a reason for this difference.

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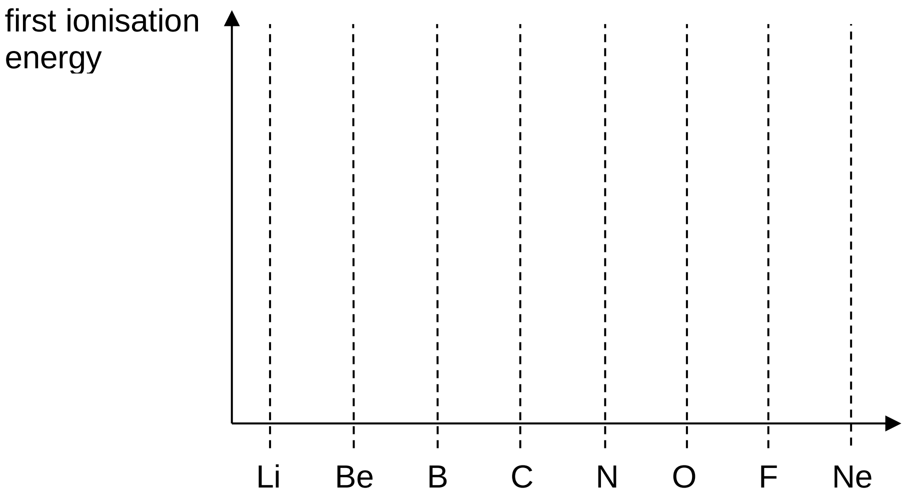
(ii) On the axes below, sketch the graph for the trend of first ionisation energy of period 2 from lithium to neon.



(iii) Explain why the first ionisation energy of oxygen is less than that of nitrogen.

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…...…………………………………………………………………………………...[4]

(d) Each molecule of propene has a C=C bond which is much stronger than a C–C bond.

(i) Draw a diagram to show the orbitals overlap for the bond that is present in C=C but not in C–C.

(ii) Draw the displayed formula of the product when propene reacts with aqueous bromine.

[2]

[Total: 13]

3 An unknown compound **A** containing C, H and C*l* was purified and subjected to elemental analysis. Combustion of a 0.200 g sample yielded 0.155 g of H_2O. Further analysis showed that the compound contained 57.4% of C by mass.

(a) Show that the empirical formula of compound **A** is C_5H_9Cl.

[3]

Compound **A** decolourised purple $KMnO_4$ solution to form effervescence.

(b) Explain the reaction described, including any possible deduction.

…………………………………………………………………………………………......

……………………………………………………………………………………………..[1]

A reacts with gaseous hydrogen chloride to form compound **B**. Compound **B** reacts with hot aqueous potassium hydroxide to form **C**.

(c) Name the type of reactions for the conversion of **A** to **B** and **B** to **C**.

Type of reaction for **A** to **B** : ………………………………….

Type of reaction for **B** to **C**: ………………………………….

[1]

When compound **C** reacts with alkaline aqueous iodine, one mole of **C** forms two moles of yellow solid.

(d) Explain the reactions described, including any possible deduction.

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..[1]

(e) Using the information from **(a)** to **(d),** draw the structural formula of **A**, **B** and **C**.

A	**B**	**C**

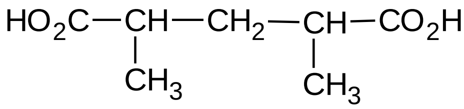
[2]

When **B** reacts with hot alcoholic potassium hydroxide, two *non-symmetrical* isomers are produced.

(f) State the type of isomerism exhibited by the products of this reaction. Draw the two isomers.

[2]

(g) Outline how 2,4-dimethylpentanedioic acid can be produced from **B**. State the reagents and conditions, and the structure of any intermediate formed.



2,4-dimethylpentanedioic acid

[3]

[Total: 13]

End of Section A

Section B : Free Response Questions

*Answer **two** of the following three questions.*
Answer these questions on separate answer papers.

4 One of the main uses of hydrocarbons is as fuels for internal combustion engines of motorised vehicles. Fuels sold commercially contain many compounds, depending on various considerations, including engine efficiency, climate and pollution concerns.

(a) One of the pollutants of internal combustion engines of motorised vehicles is nitrogen dioxide, a brown gas. Nitrogen dioxide can undergo dimerisation to form dinitrogen tetroxide.

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

(i) 1.0 mol of NO_2 is left aside in a 280 cm^3 glass jar at 25^oC. When equilibrium is established, the percentage conversion of NO_2 is found to be 82%. Calculate the equilibrium constant, K_c, for this reaction at 25^oC.

(ii) Draw the dot-and-cross diagram for NO_2 and N_2O_4. Hence, or otherwise, deduce, with reasoning, whether the dimerization reaction is endothermic or exothermic.

(iii) One of the pollution issues affecting major industrialised cities is smog, partly caused by NO_2. This often gives the atmosphere a brownish colouration.

Using *Le Chatelier's Principle* and your answer in **(a)(ii)**, explain why smog is more commonly found in warm sunny cities.

[7]

Fuels are blended to achieve different octane numbers. Octane number is a measure of the compression extent fuels can withstand before detonating in combustion engines of motorised vehicles. The compounds 2,2,4-trimethylpentane and heptane are assigned a reference octane number of 100 and 0 respectively.

(b) (i) Draw the structural formula of 2,2,4-trimethylpentane.

(ii) When 1 g of 2,2,4-trimethylpentane is burnt, the heat released raised the temperature of 240 g of water by 33 ^oC. This reaction has an efficiency of 70%. Assume that the specific heat capacity of water is 4.2 J g^{-1} K^{-1}.

Calculate the enthalpy change of combustion of 2,2,4-trimethylpentane.

[4]

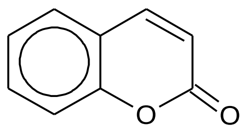
(c) 2,2,4-trimethylpentane may undergo substitution with chlorine in the presence of sunlight to form monochlorinated compounds.

(i) Draw all possible monochlorinated products. Classify them as primary, secondary or tertiary halogenoalkanes. Using alphabets, label each of your products.

(ii) If the secondary halogenoalkane is two times as easily formed and the tertiary halogenoalkane is three times as easily formed as the primary halogenoalkanes, predict the molar ratio of formation of each of the isomers that you have drawn in **(c)(i)**. Use your labels in **(c)(i)** to represent your products.

[5]

(d) Another source of fuel is kerosene, mainly used to power jet engines of aircrafts. In the United Kingdom, kerosene is dyed red. The chemical responsible for the red colouration is *coumarin*.



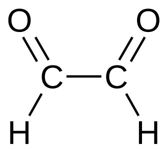
coumarin

Draw the structural formula of the organic product formed when *coumarin* undergoes prolonged heating with acidified potassium manganate(VII). [1]

(e) Methylbenzene is a fuel additive used as an octane booster. Methylbenzene can react with bromine under **two different conditions**. State the two conditions. For each condition, draw two possible organic products formed. [3]

[Total: 20]

5 Ethanedial, also known as glyoxal, is used in the production of fabrics which have permanent creases. This yellow colored liquid is the smallest dialdehyde (two aldehyde groups). Ethanedial undergoes many of the reactions of aldehydes.



ethanedial

When ethanedial is reacted with NaOH and the product treated with dilute sulfuric acid, the following reaction sequence takes place to produce glycolic acid. Glycolic acid, $CH_2(OH)CO_2H$, is an α-hydroxy acid (AHA) used in various skin-care products.

Step I $\quad$ $CHOCHO + NaOH \rightarrow HOCH_2CO_2Na$
ethanedial

Step II $\quad$ $HOCH_2CO_2Na + H^+ \rightarrow HOCH_2CO_2H + Na^+$
glycolic acid

(a) State the functional groups present in glycolic acid. [1]

(b) Describe **two** simple test-tube reactions to **positively** identify ethanadial from glycolic acid. Write balanced equations for the reactions. [6]

(c) The acid dissociation constant, K_a, for glycolic acid, $HOCH_2CO_2H$, is 1.48×10^{-4} mol dm^{-3}.

(i) Glycolic acid is a *weak Bronsted acid*. What do you understand by the terms in italics?

(ii) Write an expression for K_a of glycolic acid.

(iii) A solution of glycolic acid has a concentration of 0.20 mol dm^{-3}. Using the given concentration and K_a values, calculate the pH of this solution.

(iv) A solution containing glycolic acid and its glycolate salt acts as a *buffer solution*.

What do you understand by the term *buffer solution*? Explain, including appropriate equations, how this buffer solution works.

[7]

(d) Under appropriate conditions, glycolic acid is capable of forming a cyclic compound **W** of molecular formula $C_4H_4O_4$. Suggest the structural formula of the product **W**. [1]

(e) Ethanedial can be synthesised from ethanedioic acid, $(COOH)_2$.

(i) Suggest how the synthesis can be carried out. State the reagents and conditions, and the structure of any intermediate formed.

(ii) Explain, in terms of structure and bonding, the difference in the boiling points of ethanedioic acid and ethanedial.

[5]

[Total: 20]

6 Compounds containing the allyl group, $CH_2=CHCH_2–$, have pungent smells and are found in onions and garlic.

Allyl alcohol, $CH_2=CHCH_2OH$, is a colourless liquid that behaves as an alkene and as a primary alcohol.

(a) Suggest the structural formula of the organic compound formed when allyl alcohol:

(i) is reacted with cold alkaline sodium manganate(VII).

(ii) is heated under reflux with an acidified solution of dichromate(VI) ions.

[2]

(b) There are various methods to produce allyl alcohols.

(i) Allyl chloride can be hydrolysed with sodium hydroxide to produce allyl alcohol. Suggest the structure of allyl chloride.

(ii) Propene can undergo acetoxylation with oxygen and ethanoic acid to give allyl acetate as shown.

$$CH_2=CHCH_3 + \tfrac{1}{2}O_2 + CH_3CO_2H \rightarrow \underset{\text{allyl acetate}}{CH_2=CHCH_2OCOCH_3} + H_2O$$

State how allyl alcohol, $CH_2=CHCH_2OH$, can be produced from allyl acetate in the laboratory.

(iii) Propene can be oxidized to give *acrolein*, which upon hydrogenation gives allyl alcohol. *Acrolein* is a planar molecule which give a precipitate with alkaline copper(II) ions. Suggest a structure for *acrolein*.

(iv) Certain compounds can be oxidised by selenium dioxide to give allyl alcohol.

Selenium dioxide, showing similar behavior as sulfur dioxide, can react with a base to give salt and water. Write an ionic equation for the reaction of selenium oxide with a base. Briefly explain whether this reaction is expected of selenium dioxide.

[5]

(c) **(i)** Allyl alcohol can be converted into propanal in a single step using a ruthenium(IV) catalyst in water.

$$CH_2=CHCH_2OH \xrightarrow{\text{ruthenium(IV) catalyst}} CH_3CH_2CHO$$

The reactant and the product are isomers. State the type of isomerism displayed.

(ii) Suggest what is unusual about the single step reaction in **(c)(i)**.

[2]

Allyl alcohol, $CH_2{=}CHCH_2OH$, can react separately with sodium and chlorine, elements of Period 3.

(d) **(i)** Draw the displayed formula of the compound formed for the reaction of allyl alcohol with:

1 sodium.

2 chlorine.

(ii) State and explain for the difference in electrical conductivity between sodium and magnesium.

(iii) Describe the reactions when magnesium chloride and silicon tetrachloride are separately added to water, suggesting the pH of the resulting solutions and writing equations where appropriate. State the effect of the resulting solutions on Universal Indicator solutions.

(iv) Magnesium and sulfur can react to give magnesium sulfide, a compound with melting point above 2000°C. Magnesium sulfide reacts with water to form hydrogen sulfide.

$$MgS + 2H_2O \rightarrow H_2S + Mg(OH)_2$$

1 Draw a dot-and-cross diagram for magnesium sulfide.

2 Explain whether or not the reaction of magnesium sulfide with water is a redox reaction.

[11]

[Total: 20]

End of Section B